

EXP-11 X86 CODE

Lex Code:

```
11.1
/mnt/c/Users/mariy/OneDrive/Desktop/Compiler/EXP-11 X86 Code
Save
11.l
11.y
1 %{
2 #include<stdio.h>
3 #include<string.h>
4 #include "y.tab.h"
5 %}
6
7 %%
8 [a-z][a-z0-9]* { strcpy(yyval.var,yytext); return NAME; }
9 [+]          { strcpy(yyval.var,yytext); return PLUS; }
10 [=]         { strcpy(yyval.var,yytext); return EQ; }
11 [-]         { strcpy(yyval.var,yytext); return MINU; }
12 [*]         { strcpy(yyval.var,yytext); return MULTI; }
13 [/]         { strcpy(yyval.var,yytext); return DIVI; }
14 \n         { return '\n'; }
15 [ \t]       { /* ignore */ }
16 %%
17
18
```

Yacc Code:

```
1 %{
2 #include<stdio.h>
3 #include<string.h>
4 #include<ctype.h>
5
6 FILE *fout;
7 int yylex();
8 void yyerror();
9
10 %}
11
12 %token<var> NAME PLUS MINU EQ MULTI DIVI
13 %type<var> exp
14 %union{ char var[10]; }
15
16 %right EQ
17 %left PLUS MINU
18 %left MULTI DIVI
19
20 %%
21
22 input : line '\n' input
23       | '\n' input
24       | /*empty*/
25       ;
26
27 line : NAME EQ exp {fprintf(fout,"STA %s\n",$1);}
28       ;
29
30 exp : NAME PLUS NAME { fprintf(fout,"LDA %s\nLDT %s\nADDR A,T\n",$1,$3); strcpy($$, $1); }
31     | NAME MINU NAME { fprintf(fout,"LDA %s\nLDT %s\nSUBR A,T\n",$1,$3); strcpy($$, $1); }
32     | NAME MULTI NAME { fprintf(fout,"LDA %s\nLDT %s\nMULR A,T\n",$1,$3); strcpy($$, $1); }
33     | NAME DIVI NAME { fprintf(fout,"LDA %s\nLDT %s\nDIVR A,T\n",$1,$3); strcpy($$, $1); }
34     | NAME { fprintf(fout,"LDA %s\n",$1); strcpy($$, $1); }
35     ;
36
37 %%
38
39 void yyerror()
40 {
41     printf("\nError ");
42 }
```

```

21
22 input : line '\n' input
23 | '\n' input
24 | /*empty*/
25 ;
26
27 line : NAME EQ exp {fprintf(fout,"STA %s\n",$1);}
28 ;
29
30 exp : NAME PLUS NAME { fprintf(fout,"LDA %s\nLDI %s\nADDR A,T\n",$1,$3); strcpy($$, $1); }
31 | NAME MINUS NAME { fprintf(fout,"LDA %s\nLDI %s\nSUBR A,T\n",$1,$3); strcpy($$, $1); }
32 | NAME MULTI NAME { fprintf(fout,"LDA %s\nLDI %s\nMULR A,T\n",$1,$3); strcpy($$, $1); }
33 | NAME DIVI NAME { fprintf(fout,"LDA %s\nLDI %s\nDIVR A,T\n",$1,$3); strcpy($$, $1); }
34 | NAME
35 { fprintf(fout,"LDA %s\n",$1); strcpy($$, $1); }
36 ;
37
38
39 void yyerror()
40 {
41 printf("\nError ");
42 }
43
44 int yywrap()
45 {
46 return 1;
47 }
48
49 extern FILE *yyin;
50
51 int main()
52 {
53 FILE *fin;
54 fin=fopen("input.txt","r");
55 fout=fopen("out.txt","w");
56 yyin=fin;
57 yyparse();
58 fclose(fin);
59 fclose(fout);
60 return 0;
61 }
62
63
64

```

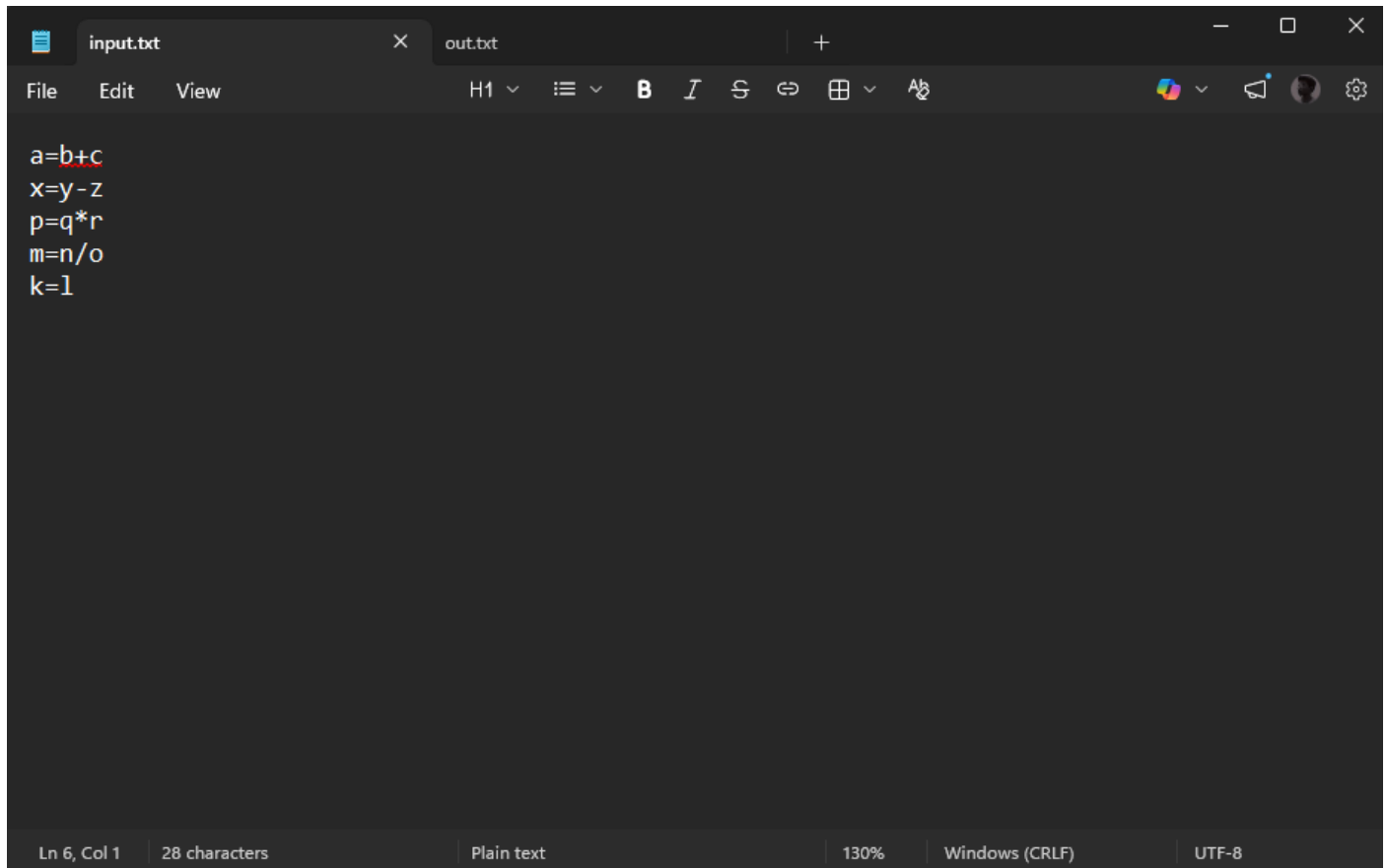
Outputs:

```

mariy@Venkata-HP-Touch: /n x + v
mariy@Venkata-HP-Touch: /mnt/c/Users/mariy/OneDrive/Desktop/Compiler/EXP-11 X86 Code$ lex 11.l
mariy@Venkata-HP-Touch: /mnt/c/Users/mariy/OneDrive/Desktop/Compiler/EXP-11 X86 Code$ yacc -d 11.y
mariy@Venkata-HP-Touch: /mnt/c/Users/mariy/OneDrive/Desktop/Compiler/EXP-11 X86 Code$ gcc lex.yy.c y.tab.c
mariy@Venkata-HP-Touch: /mnt/c/Users/mariy/OneDrive/Desktop/Compiler/EXP-11 X86 Code$ ./a.out
mariy@Venkata-HP-Touch: /mnt/c/Users/mariy/OneDrive/Desktop/Compiler/EXP-11 X86 Code$ |

```

Input File:

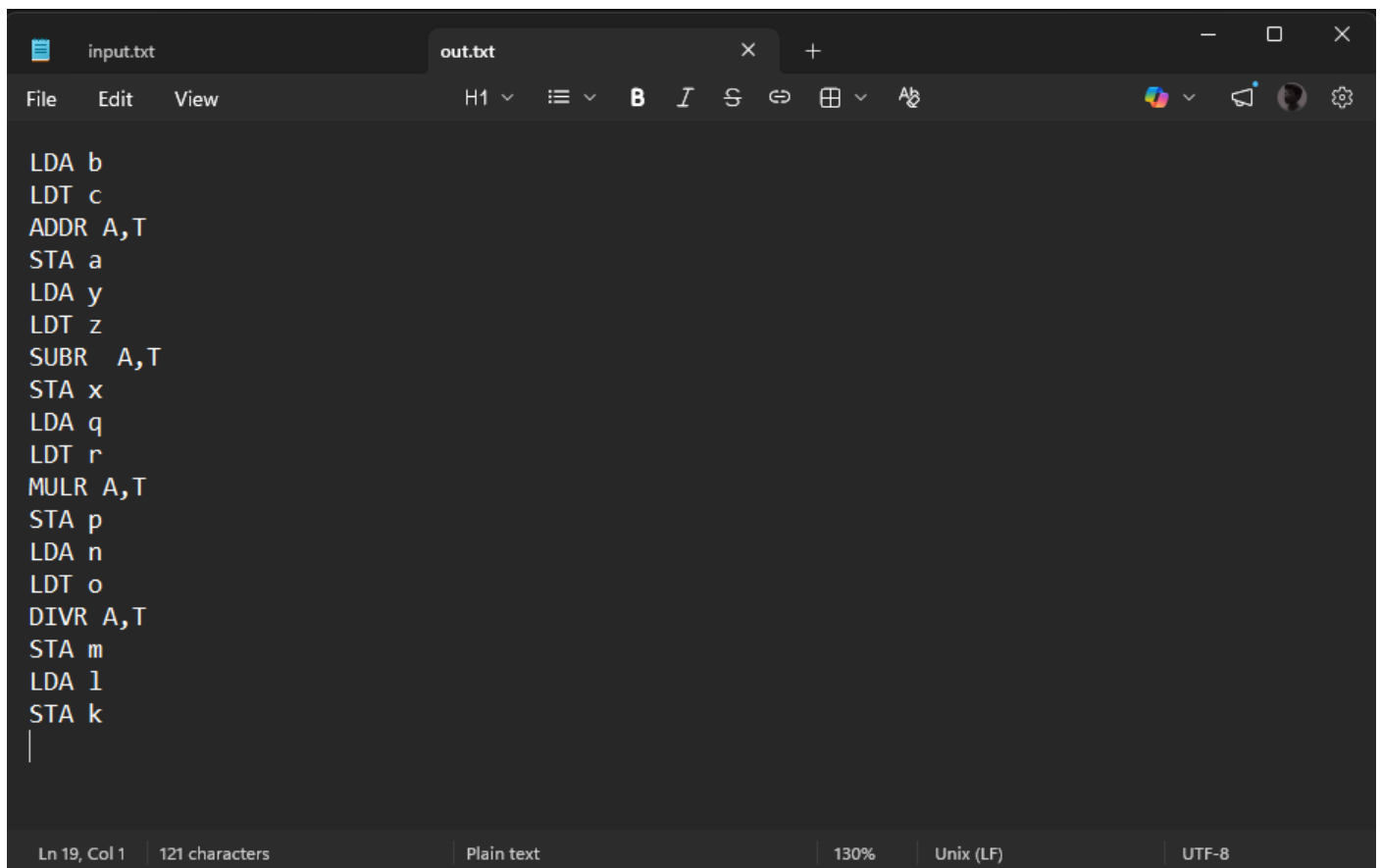


A screenshot of a code editor window with two tabs: 'input.txt' (active) and 'out.txt'. The editor has a dark theme and a menu bar with 'File', 'Edit', and 'View'. The toolbar includes icons for font size (H1), list, bold (B), italic (I), strikethrough (ABC), undo (↶), redo (↷), grid, and a keyboard icon. The main text area contains the following code:

```
a=b+c  
x=y-z  
p=q*r  
m=n/o  
k=1
```

The status bar at the bottom shows: 'Ln 6, Col 1', '28 characters', 'Plain text', '130%', 'Windows (CRLF)', and 'UTF-8'.

Output File:



A screenshot of a code editor window with two tabs: 'input.txt' and 'out.txt' (active). The editor has a dark theme and a menu bar with 'File', 'Edit', and 'View'. The toolbar includes icons for font size (H1), list, bold (B), italic (I), strikethrough (ABC), undo (↶), redo (↷), grid, and a keyboard icon. The main text area contains the following assembly code:

```
LDA b  
LDT c  
ADDR A,T  
STA a  
LDA y  
LDT z  
SUBR A,T  
STA x  
LDA q  
LDT r  
MULR A,T  
STA p  
LDA n  
LDT o  
DIVR A,T  
STA m  
LDA l  
STA k  
|
```

The status bar at the bottom shows: 'Ln 19, Col 1', '121 characters', 'Plain text', '130%', 'Unix (LF)', and 'UTF-8'.